

ActInf GuestStream 061.1 ~ Shagor Rahman: "Myth of objectivity and the origin of symbols"

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hello and welcome everyone to active inference Institute it is October 20th 2023 and we're here in active guest stream 61.1 with shagor Ramen so today we'll be having a presentation myth of objectivity and the origin of symbols followed by a discussion so thank you very much for joining looking forward to the talk thank you very much s for coordinating this um and and thanks yeah for all the the work that you and the active in active inference Institute does um also wanted to briefly thank um Carl fren and Paul B Badcock for um helping me with this paper and kind of making it a reality so this is um the title of the paper here so it's called myth of objectivity in the origin of symbols um and it's in Frontiers and sociology um and so just get right into it as um as Daniel said and do you hopefully a brief overview and we can do a discussion at at the tail end this um and so I guess briefly the thesis that I'm trying to argue for uh is that and you know some of these words will need clarification um particularly in in the context we use him um but it is that the belief in Universal and kind of a codified egalitarian um social norms or in other words morality this essentially underwrites broadly our use of symbols um so what is that kind of mean functionally in terms of what other disciplines use um or how they would think of our use of symbols um well there's cultural Niche um and this you can think of as you know our ability to engage and cultural Evolution um from Joseph enrik and others who have highlighted that as an area of an important area of Humanity's distinction another one is social reality and this is um more in the psychological realm um but you know how we use things like money um you know how we um obey someone with a crown on their head um so just things that are socially constructed um realities are socially constructed as opposed to you know existing in more of a physical sense that other um non-human animals do and of course this is gets to our symbolic capacity so this is you know our symbolic Consciousness our um ability to kind of think deeply in the past to plan deeply in the future um and to engage with kind of any number of um Concepts as well as people uh and so the claim we can divide up this claim into kind of three areas um the first is our kind of

modern social dynamics and social bias and so you know how this is the sociology side um so you know how our economic systems um look how our Civic uh and political institutions look um and kind of building on um that so I think once we have the structure on hopefully I convince you of um a way we look at um Humanity's uh structure of their social reality and know their kind of symbolic reality um then you know the question is well how did we get there you know how did we become um the kind of you know primate that points fingers and feels guilt as you can kind of see in that picture um and for this we can look to you know two areas so one is you know non-human social animals so pites um you know cations and another animals that engage in social behavior of a similar nature um so this is distinct right from usocial insects from Mano ants and bees um but rather this is you know chimps um you know bonobos marasa monkeys and and some others that have been studied um specifically you know by researchers um seeking to examine Humanity social norms and how it differs from you know primat and other animals and and kind of lastly is is the are of of anthropology um so whatever we're able to you know determine about our you know the social environments of our ancestors um you know what can that tell us about how we got here and I'll make some kind of specific claims in regards um yeah and so so I set out my goal and essentially my goal is to kind of tackle that first part and to you know persuade folks um you know on the call here that our social system takes on a particular structure and that structure is a essentially a symbolic self and I use that in in the sense of carlong meaning that we inhabit um in addition to our own identities we conceive of ourselves our self-conception uh is attached to a number of other um broader cultural identities and these these could be a this could be um you know your country this could be your company um this could be your community you got to think of you those those 5c so country company um Clan Etc um and so that's essentially where that star is that that's where I'd like to get to um by the end of today and I'll also get into some of the other areas as well um and so to to explain briefly what I mean by the myth of of objectivity uh so first of all um by myth I I mean to say that we have it's not that we've obviously identified some kind of underlying structure with you know reality or with our social systems um that our symbols represent right so think here of um Plato's forms that you know there's um you know a number of chairs out there but there's a kind of an ideal form of chair um you know that's not the claim I'm making one way the one way or the other um so instead what I'm saying is Well during the um past scene we were just like another primate um trying to solve primate problems um and the way we did that or one way we did that um effectively um was through this was through what constituted essentially a myth was

essentially um establishing um you know a common reference frame um for behavior and I'll get into that term in a bit in a second and the second um piece of this the objectivity so that's to mean that it's not simply um it's not simply a claim that you know our ancestors um you know listen to stories and you know believed in myth and believed in religion um but that our myths that were powerful um were practical um Were Meant to solve um social problems and do so in um an objective way so be essentially more scientific um so one of the claims that I'm making is that our ancestors kind of similar to how we are today um or sorry similar to how some of us are um are interested were interested um in identifying um actual real world consequences um in a symbolic manner so having some sort of objective truth um and kind of you know believing that an objective truth exists um so kind of right and wrong exist in an objective way um could in bad exist in the sense of um you know this can kill me this this can't kill me um and objective in the sense of you know this is part of my in group and this is outside of it all right so roughly how did we get there um so more specific um on the uh anthropological side um um I'm G to look at things like shth um so that this is our most innate social bias um which is important because um for a few reasons but one is that um Society these social social units even outside of the human Spectrum um some researchers have used um things such as um social bias to analyze um the structure of non-human animal social units um so essentially in group preferences um social biases in that respect um are ways that we can understand how other animals how chimps and bonobos interact with one another how they establish friends and um and distinguish between VES and essentially how they exist in these um you know normative um you know normative social units and hence our use of shth speaks to kind of a unique way that Humanity um expanded upon our social biases and so although it's you know a Crux now um you know it's it's often seen as problematic um there's a certain contingency of folk that c should as um a necessary step in our Evolution um so I adopt that view and um see that as a an important kind of factor in our road to to morality right um and so within that kind of shth environment um um which is essentially we can think of as still non symbolic um because shth um as um I should probably should probably explain what should with this um so this is essentially accent bias um so we're kind of taken from um the Old Testament but it represents uh our kind of innate tendency um to associate and to establish an instinctive kind of common ground with folks who sound like us um and so we you know automatically trust folks um you know who grew up near us who have a similar accent um and the further out we get um often times um the more foreign some can can see not always the case but kind of instinctually and you know prior to the acquisition of symbolic identity we'll

use sh will will typically um you know practice shth as a way to to um identify and so you know assuming that this that this practice of shib withth you know goes kind of far back into our past um then you know how do we transition out of that um to something more symbolic you know how do we you know start using morality how do we start using language um well here our uh my suggestion is that we use one of or one way to think of it is using um what's called a speech act and so this comes out of the um the field of the philosophy of language John Surly and and Austin a bit earlier than that um but there's speech acts called performative and constitutive utterances and these are essentially acts that um immediately change uh the environment in some way and so um if I suddenly tell you you know that you're fired um or that um you know um um you know there's a there's a war um across the seas right now um that has immediate consequences even though it was just a speed check so before any violence happens before um someone is now kind of crowned a king and starts implementing physical policies just by speaking you've changed something um and so in this kind of examples my kind of stylized example is that um well someone committed um some kind of Act of deviation they violated some Norm um and so some vocalization was emitted that um labeled that person so this person should have been an ingroup member this person kind of should have been aligned with our in group's Norms but they violated in some way and it could have been you know they violated food sharing they they violated um you know some of Unwritten contract about Cooperative breeding like how we that we should protect um you know infants um um and it could be done a number of different normal violations but labeling that person um you know as a deviant would constitute kind of a minimal threshold of a speech act um that could be read as symbolic and so the the starting off point here actually is um or one of the important starting off points here actually um is what is itself s and so um you know symbols are are one thing and um we can probably identify that and and have some discussion but we have some kind of clear idea of what the self is um sorry what a symbol is it turns out however that you know defining the self is a bit um it's it's not as confirmed and locked in um as at least I would have expected kind of when I when I started this um this project and so typically agents are um defined we get we'll get to that in a second they're they're U they're defined as along their person um but it's it's not clear where um say a cell starts where um an organ starts um you know where a person starts and where Society is um and so what we'll get out here is um using some Frameworks we'll see how we can understand bias perspective shared go goals reference frames um and the next topic which is um the free energy principle see how we can use those to start understanding um you know how self and how identity how

how we can think of um the self and identity in the context of um you know our creation of symbols right and so yeah roughly I mentioned some of these already but um what we're going to look at is non-human sociality um we look at you know religion myth narrative this is you know of course a very instinctual very ancient um practices um that defines Humanity um we'll get a social bias as I mentioned um and then ancient sh and most importantly you know this is the active inference Institute um and so the framework that I'll U be positioning this within um is the free energy principle and more specifically Act of inference yeah just just one more over um so again fren principle um we'll go over the ideas of identity uh and how we can start to think about this subject and how that is evolving um in the light of these new Frameworks of the fre in principal active inference and some other researchers um who are sympathetic um to these views uh and then the the weird journey and I'll explain that's an acronym so I'll explain what that acronym is uh in a bit but essentially for now it's just you know Western Society it's so Western Society is zone for being particularly analytical um and as a result of that one of the consequences of that is um most recently and this is you know around the past Century so around 20th century a lot of um disciplines emerged um that did something um that had a common approach um which was an attempt to um you know reduce uncertainty um identify agents um and you know um provide kind of explanatory Frameworks in the field of um that first one post structural this is in the field of linguistics the second one of course is economics um and the third one is behaviorism in the field of psychology and so we'll just kind of go through those and see what we can learn from um these paradigms um when we try when I try to essentially you know convince and and um know discuss Humanity as um you know an agent that occupies a symbolic self yeah so in addition to of going the route that um I mentioned so far we'll also discuss what this means for moral philosophy um and you of course this is we we'll just um kind of dip our toes in we won't go too far into moral philosophy but I think the implications are kind of immediately um clear in some areas um it brings some interesting ideas from the past of you know what is a bit more relevant um what's a bit less relevant um whether you know effective altruism and things like that are kind of consistent with this view or kind of whether they're um at odds um and the last piece if we have time just because uh I know there's a similar stream on um Cooperative communication so um I'll kind of give my view on how this um kind of aligns with Cooperative communication and um communication viewed as kind of an ecological framework which you know both of those were um purportedly done within the active inflence framework so yeah and so uh to start things off um the free engy principle so just kind give a brief explanation so

is the active infer Institute um so there's obviously you know a great kind of wealth of uh information um that explains the Paradigm and how it's useful and some of the terms but essentially it's a it's a framework that essential that um attempts to identify um a number of formalisms um that describe the science of self-organization or autopi posis and it represents uh agents things objects um everything as essentially statistical models of their environment uh and so more specifically a agents need to possess a um that minimum uh they need to possess a generative model of their environment in which they can um understand in a sense or infer the processes of the world so you know how does a um you know a creature navigate his environment how does an an um you know find its way back home they have the they themselves rep are can be represented as models of their environment um so we can see that as as uh these agents encoding basan beliefs um and um so one way to look at and this isn't part of the formalism that's why it's tasz um but I I heard Chris Fields say it's all I'll say it here which is that one way to think of it is you know agents are essentially all scientists they have beliefs about the world um they act on the world in a certain sense um and those beliefs get updated so everyone has a running hypothesis of what they should next do who they are and how they should maintain themselves their environment and so at a basic level you know everything that exists in the world has to engage in in some sort of Science and this uh occurs at various scales um so this is at cells at organs um you know agents people um or animals of you will societies and and culture and that kind of last area of course is where um our focus is here today and this is a tal size as well but um one of the takeaways from the free Ander principle is that a foundational element is the search for oneself and so it's not um a given that we understand precisely who we are um in the world and it kind of extends to who your friends are um and who your enemies are and um and um but this surprise imization uh can be thought of as this um Socratic search for self you know meaning you know Socratic Socrates victim to to know thyself um there's something ancient there and not just um you know for moral philosophy yeah and and one other point to make is that um although belief uh in the FY principal context is um a technical term it's it's refers to um basian statistics it does actually come out of um a use at a higher level so Ed Thomas spay was a preacher and he was specifically talking about religious Faith um and it was a formal and he made a formalization for how we he could think about um his religious beliefs and so that's ultimately what led to um basian statistics um and so one attempt or one um kind of takeaway I I I have is that we can think of due to um principal scale and variant nature we can think of um beliefs that's we typically use the term the religious context and and another context um we can you know try

to formalize these within the free energy principle um not as a given um you know we could get things wrong um but it's at least conceivable that they're contained within that all right the next area so in addition to um the free energy principle there's of course other work that's been done to understand identity understand self um and whether that exists kind of at the bounds of the skin or whether that exist you within social units um and so one interesting um U one interesting piece that's been done um is by Michel Lan who focuses on bioelectricity at the cellular level and how cells communicate and there he actually used Game Theory which is a paradigm typically used in um in economics but he demonstrated how uh cells if we can think if we can model them not just as kind of individuated um little agents trying to maximize their own interest if we can set mod of them as potentially inhabiting um a wider self we can see how cooperation would emerge um in a way that um more competitive paradigms more reductive paradigms would die and so immediately even before we get to how Humanity acts with other humans um whether we do that cooperatively whether we use morality with use other Norms um the same problem exists you know at the level of the cell um because and the second piece is once we get to um once we get to chimpanzees if we're over in the the upper right hand corner um what we uh what we understand there is um for Mark Moffett Sanz determine their group size by social intimacy and so this means that their group size essentially cannot exceed generally does not exceed um what any individual chimp is able to recognize so they basically have to re so each person each member of the group has to recognize everyone within that as opposed to you know as opposed to us and humans um which the already use has existed for you know far into our past we've been in Anonymous social units right um and so here um what I did was before I get to um further down the line um what I did was take what I I took another concept of Michael Evans which is this light cone so um the light cone essentially is one way of representing identity and so the the scope of each agent goals um can be kind of captured in this idea of like cone and so in in the area of a chimpanzee there their goals are associated with others in their that that cohabitate in their social unit um and they're necessarily subjective and so they're necessarily um based on their own perspective right and so another piece of social identity another way to look at it um and so there's a debate whether or not um we can model animals as being in societies and so this isn't a given um and so there's you been arguments between um you know dawkins and um Edward Wilson for instance that you know claim that e side kind of claiming that you can you either can um think of animals as um existing in Social units and those operate similar to cells or they can't one way around that um on this point number three researchers found that you

can model them actually as just possessing ingroup favoritism so if instead of kind of worrying about specifically where the boundary is from our point of view from um you know an outsider's point of view if you inste model it as well each individual agent um so like I said Each one um maintains uh an intimate recognition of others within their in group um following that they show preference refences to those inside so this could be as simple as when there's a fight when there's an attack from an outside group you'll prefer um those within your ingroup versus out group um it could be food sharing it could be hunting and things like that but it doesn't have to be extensive cooperation it could be simple um preferences it could be avoiding spending more time around your in group versus um outside your outside group and so this is one of the ways in which um social identity um you know exists at this at this level and so another important piece is um which Judith burkart um in K's measured which was implicit Norms uh and subjective Norms that that are exhibited by these chimpan and so as an example if chimp saw um an in group infant being harmed um they would experience um a visceral reaction um they would have heightened Razzles they would pay attention um and so this kind of suggests that there are norms that chimpanzees and other animals um and other animals exhibit and some animals even exhibit a more extensive um more extensive Norms such as marasa monkeys um but chimps are on the lower they still they're on the lower end of that Spectrum but they still exhibit norms and so one way that um Judith Bart distinguished between um Prime immorality and human morality um was to say that we have a top- down third party perspective um and so this is where I get the goofy um graphic where if there's a triangle on top of um each of these each of one of our ancestors heads um is because our ability to engage in objective morality um can be represented by this shift between um perpendicular uh perspective perspective where I'm looking at those around me and and I'm imagining them looking back at me versus I'm taking a top down perspective uh and evaluating my actions and the actions of others in the context of some more transcendental perspective and so this and that object that view of objective morality um versus subjective um social norms we'll say was also brought up by Beck off and Pierce um they did claim that morality exists in other animals but they were using Morality In A broader sense non symbolic way um and they make a lot of good points with this is and they go across wolves they go across um crustations you know lobsters Etc um how these animals engage in essentially behaviors that suggest Norms but also exhibit behaviors that suggest emotions that are underly that we typically think as underlying um Norms of morality and so the last piece I'll mention is that um morality for for humans is typically um is typically symbolic is pretty strongly

symbolic and the work of Shiner and others will typically or have found that in order to identify an action or a behavior as being a moral constituting an immoral behavior it'll typically need to violate some principle or some code so continuing on the idea of what establish as a subject you know how do we first identify what an individual is and what can this tell us about how we manage something like a symbolic self and how does a symbolic self then become to be constituted by you know by a belief in Morality or what is good for the overall self versus the individual the work of Markus who used active inference and the free energy principle to understand the function emotions and what his one conclusion he had is that emotions is the the spot of of Consciousness and that should be the direction of consciousness research so I won't go that far but one point that he makes that's relevant here is that the phenomenon of emotion is relevant specifically in helping manage categorically distinct needs so distinguishing between you know pain and outside pain and hunger which one's more important do I go to sleep or do I have food managing these various you know different substance systems is the function of emotion and can start to help start to help us understand the you know the boundaries of the self and so if we go back to thinking about how Miller looked at an agent how these things comes together we can see how emotions fits nicely into that picture because it helps essentially harmonize these various disparate systems into a person right this so get this gets at hedonic valence this is kind of pleasure versus displeasure so this is an important Dimension there's a few Dimensions that researchers have identified you know with regards to emotions so you know how do we study emotions in in a rigorous fashion valence of course is a important measure there so kind of pleasure versus displeasure you can think of it as pain but typically it's this understood specifically as pleasure versus displeasure and it's in that sense that complex organisms start to move and help navigate the world and on so social creatures how they navigate the world you can see too how it could help us conceive ourselves as within the symbolic self and so some research on the idea as a symbolic self so what I'm hoping here is that my idea here is that although this was brought up by previous psychologist at earlier times I think it's clear that you know thinking of subjectivity and thinking of uh symbols in the context of active inference and you know minimizing uncertainty and a more expansive view of the self the Frameworks that that I list here can become more operationalized so Carl Jung found that you know this the self as symbol or the cosmic man was a motif that existed in numerous cultures from you know Hinduism to the ancient Greeks you know to ancient China there was is this

conin theme of the symbolic man and so each person's part from that view um you know from a religious persective from theth mythological perspective um everyone was see conceived as part of a common person so this helped um establish essentially a common reference frame for behavior um so this is and and similarly Victor Frankle Locos of therapy um who's a holocaust Survivor and also a psychologist who emphasize the need for humans to find meaning in groups and ultimately find a meaning um in lives and our well-being is was largely determined by that um yeah just kind of broadly um our sociological background so in a sense we do a number of things um that we kind of take for granted but that are all essentially symbolic and that all you represent our um you know ability to inhabit not just our own identities um but we attach ourselves with cultural with cultures uh that are in some cases extinct and in the past with Futures that we kind of have youf to um understand so I do want to and discussions think I'll get to the to the final thesis with um but I I would like to kind of briefly go over um the weird Journey that I have here um and so uh I think this demonstrates how a couple things um and but analyzing um Western Sciences so Western science uh paradigms in the field of economics and and language and behaviorism can help um bring some of the the points I'm making to light and so we are the acronym that I mentioned earlier this is um Joseph enri um it's Western educated IND industri highs rich and Democratic uh as you can see so essentially what he found was a lot of uh scientific studies relied on college students um they were typically college students in in wealthy countries and they were well of course well educated because they're in college um and this caused a lot of research to have certain biases on that if you start including other um other societies that are less um weird um hunter gatherers for instance Etc you get different results so he came up with this acronym um but then he he doubled down on it and in a more recent book he explained that um weird Psychology was the reason for the Industrial Revolution um the reason for the scientific um mindedness that that we exhibit um and Etc and so this and we differ and the consequences of that is we possess an analytical mind so you know think of you know a scientific oriented mind um um you think of someone who's bookish um versus more Galt um thinking which is present in more feudal and cast and clan-based um societies and institutions um so thisal thinking had you know has led more recently and it was a um a discussion a couple days ago on on using narrative Theory and economics um so that's firmly in this um you know in the message I'm trying to um try to overview here but of course in neoclassical economics is an example of a more analytically oriented but something that is essentially reductive and structuralism and behaviorism follow along the same kind of heels and so what these three U what these three paradigms have in

common is that they attempt to explain a complex phenomenon by um reducing uncertainty although minimizing uncertainty is important within the F principle it's also important to um understand that it's that it's minimized and that minimization happens um you know dynamically within systems and is part of the you know adaptive process um and essentially how we get the systems that we do and how they are Dynamic how they change over time um how we have preferences um Etc and so so one question is well if if these um you know if these paradigms were overly reductive uh and don't do a great job of of you know explaining the systems that they purport to then why do we use them one reason is that they are great null hypothesis um um you know as we'll see a lot of economists like tasan neoclassical economics a lot of post structuralists like to bash on structuralists um and that kind of goes on and and you know there's good reasons for that because you know sometimes when things are bad they're they're um good at help elucidating important Concepts that are true okay and the second reason that I I bring it up today is that um they reveal moral conception at the heart um of how Humanity deviates how our social systems deviate from these predictions um and a third reason is that the hund gatherers were also analytical um so although they differed from us in important ways one thing they shared um from us was an analytically oriented mind so that means that the clan-based and and the feudal institutions that Henrik was talking about this happened only past the Agricultural Revolution so this was 10,000 years ago um when most societies essentially shifted their normative structure became more stationary and as a result of that adopted the these more GTI views one way to look at um our trajectory two symbols was well our our um you know our ancestral past um was filled with people trying to do science using symbols they trying to categorize um plants and animals hunting methods so anything they used for cultural Evolution and most importantly they try to classify one another they try to type one another um I'll do one example um from that and then I'll I'll move it over to a discussion so there's a few um areas that uh I wasn't able to get to but I'm have to do that as a follow-up and then I think most importantly it be great to get some questions in um but yes there's the idea of transcendental signification so this is one example um where the poststructuralist the philosopher Derrida and P had a similar critique um of what was known as structuralism so for some background Claude Levi Strauss around 1960s um maybe a little bit before that ideology known structuralism which attempts to view cultures view language and some other um institutions that he examined as corresponding to an underlying and stationary structure and so it's removing the variability from different cultures from different myths and believing that there's a certain structure to this and Derrida and other's critique of this was well

essentially we're drawing arbitrary lines and culture isn't so easily um reduced to a specific culture um there's time there's history there's of cultural variation um and one example is um EV trying to draw a hard line between nature and culture when really that those things are continuous in a different cultural a different natural environment you might say um you'll have a different culture um and the culture will feed off of that natural environment and so as darus there's nothing outside the text so everything is a matter of perspective um in that sense so the question then becomes well if uh if this if that static structure that CL Levi stous um purported existed then does do cultures have any structure and of course DED and others still did believe that there is room for a structural science right and so the structural science we're looking at today of course is the free and principal on active inference um and so what is the kind of core components what is that um I didn't go over the concept um today but where is that you know Markov um blanket or boundary that we're minimizing uncertainty within um and that's where the transcendental signification um idea comes in and this where said well there's typically especially in Western societies um there's typically a concept that you just don't that you can't argue against that's just something accepted um and from there that the constant daisy chain of Concepts the constant flow of Concepts can stop essentially and so you think of any um at any given time it took me a minute to think about what our transcendental signification is today um you know because there's a lot of fluid concept and there's a lot of arguments um but one examp example is freedom so basically at least so Ian so if someone brings up in a political context or certain other context that this will infringe upon Freedom um then it's very hard to argue against that and that's because it's it goes without saying that um you know although there could be in some context infringements on freedom it's generally understood that you know freedom is a good thing it's a positive thing um any kind of disagreements from it will have to begin with that rightness and so although D was critical U Western society's use of transational significations and um you know that's one modern example but of course in others are you know the belief in God um democracy other things that are just taken for granted that um the culture should abide by um and although he was critical of these or our use of them without transcendental signification is difficult to imagine how a culture would be maintained I think it's also difficult to imagine how you know language would um remain stable that's because there needs to be some agreement um there needs to be essentially some common ground in order for communication to happen um that's because of course language involves adopting a whole bunch Norms um but also our daily lives that are kind of empowered by language and symbols um need some sort of um of stopping point need some sort of um

ground to say well you know this this isn't a good thing or this is a good thing we shall kind of Orient ourselves to that um and so this is one representation of um why moral conceptions are um kind of necessary for our systems um and you can see that in language and you can see that I think in um in economics and the field of social trust as well um that I can go into that kind of later time um yeah the last thing is um this relates as similar things happen with the motion but I'll stop there actually and um you know folks want to do kind of a followup we can we can do that but that's want to turn on see if there's any uh questions at the moment so hey Daniel wow well awesome a lot of uh pieces arising so for those who are watching live they can write questions in live chat and I wrote down a few things as well so just while we get started I mean what brought you to bring in active inference and we explored the role in the coming questions but what what brought you to this were you studying theories of cognition and looking for what or how did it come to be this way so I guess techic I took the high road I took the high road to inference so I um I was studying economics and I thought to myself well we have economics has a behavioral framework um it might not be great but it's it's related to um utility it's related to welfare it should actually be a pretty good starting point for understanding morality so essentially it could be a utilitarian framework um except that we have prices so we should be able to clearly at least propose you know a set of um you know policies and and set of behaviors or a set of or an essentially a normative theory that you can score Behavior against to say whether it's moral or immoral um so that became very hard because it turns out that um the the neoclassical Paradigm doesn't allow that because it um it fixes individuals uh and their and their preferences um and so uh I then kind of turned to um some other moral paradigms that that used actually prisoners dilemma um and others that brought in Morality but then yeah then then I saw the um this the Scientific American or the wired article about Carl friston and him talking about the free engry principle um yeah and it seemed like okay there's there's an option here there's um you know a way to you know use math like economists love to do um but kind of in a way that still allows for um you know true subjectivity and kind of true you know variation um without kind of reducing that all the way but yeah in a nutshell I guess I I I really like this never never even heard it that way like math with a true subjectivity it's truly about the subject and what is subjective to them a lot of ways to go from there um okay you brought up shth and syntax and shith at least in its original um biblical style it had to do with how the word was said so it it was truly a question of of the morology of the word it's not like it was an unknown word it was how it was said not the underlying term um and then you seem to suggest that we were in a new phase or mode of

semantic communication that there would be something past or different than the shth so what what was that pointing to or what do you see there yeah sure I um I definitely ran through the word sh I didn't get to the um to the um anthropological side yeah so so essentially the what I'm referring to here is um I guess he had two things so one is Emily Cohen um has a paper called the case for accents um Mark Moffett has a has a book called human swarm and he kind of Echoes that that same argument there um and the last piece that I'll mention is cat katherin Kinsler um has a book called How you say it um and they're all about shith um the first two were about making the argument that um what we call shith and and what I mean I'm using it here in the context of it's just accent bias right so that in that case the um the eedes you know they or one of them killed the other one because they pronounced shth incorrectly they said eth instead of shth I think um but the point is that um you know from the second where we're in utero um we listen before we see anything um we listen to our cons specifics before we see them and there's a lot of evidence that we show preferences immediately for people who sound like the people that we heard initially sound um so we establish these a kind of orbit around our Social Circle um using that familiarity in sound um and a lot of kind of racism and a lot of other social bias can actually be deconstructed into relating either more to um shth or the accents are actually a bigger driver than we realize and we consciously realize um and so Emily Cohen um uh and Mark Moffett they use that as a way to what if we had accents kind of before we had words um and how would that how would look and the idea here is that well other animals have vocalizations um you know dog bark um chimpanzees um make vocal vocalized sounds verit monkeys they do these up and Below calls so they have in some cases sophisticated sophisticated vocalizations obviously parrots have very sophisticated vocalizations only next to us um and so would we would there be a circumstance where uh our answers are could have used that um to identify with one another um in a more cultural evolutionary perspective so why do we do it as as babies why do we do it as as children um well the argument well that probably is indicative of something that's more evolutionary ancient that has more has more legs to it um and the reason why that's important is because you know we we started off as a m primate we somehow um you know did you know defeated the chimps and the monops in terms of resources or whatever um and you know how did we start to kind of cooperate at scale um and you know there's been some discussion of that but introducing ship with clo is a lot of gaps for instance in Joseph henrik's um um you know the story of our success um or in his first book on car Revolution he goes through how [Music] we BEC how we became a species that learned from one another um and he said he admits that majority of

this happened when we live close close together but if you think about it well we must have learned from each other while we were still in these sparsely commit communities we we transitioned from these sparely communities to these higher dense environments but um it'd be very difficult to imagine a densely populated um you know area of chimpanzee because there would be a lot of fights so how do we kind of overcome that and Sh is one example by kind of extending our our our networks but yeah it's a long way to answer to your question interesting so kind of the historical and developmental Primacy of accent and then the kind of linguistic and symbolic signification that we have as a scalable technology that's why seemingly an open-ended number of people can have a word that they might all pronounce differently um or uh simple abstract simple like the triangle that you showed that that might mean something different to everybody but it's like this and and that's where you kind of invoked the platonic concept which is like the word separate from any accent like does the logos have an accent exactly exactly yeah yeah I was try to I spent like a lot of time on the the the image because uh I want to be careful because it's not that there's one um there's one light that we're all accessing right it's not that there's one perspective true perspective as far as we know that we're all that you know that Daniel's accessing when when you talk to me and that I'm accessing when I talk to you um we just have kind of we both have uh a technically Transcendent perspective that sufficiently overlaps to the extent that our communication now is is what you might call productive right to the thing that we can um properly communicate um we're co-constructing that little triangle blimp um which is kind of yeah modifi modification of of Michael Evans kind of light yeah accent encoding in the production of speech accent decoding in the recognition of speech and then that's what allows for example token based language models to be in our cognitive ecosystem and get some meanings or you know allow them to be projected meaning upon but but cool um Okay so so where does active inference come into this for you like in the paper and in the presentation you mentioned a few places but you brought in so many historical Frameworks and different directions so just in your own work leading up to here or from where you take it like what do you see active inference in the free energy principle doing in that line I I I see it as so I I I see it as operationalizing um the Y notion of you know symbol symbol AS self um and I I think that does uh a lot and um it does a lot because it gets at that uh foundational one of the foundational principles which is um seeking self so uh it's not that you know we're born and you know some children decide you know I'm not going to speak um you know I'm not I'm not super interested in in this or I I I find that I belong in this family they give me food and I'm just going to stay here forever those

kind of things don't happen um and so what I what it does for me is it's it frames the motivation um for symbolic communication um because if it's conceived of as an extension of your identity um if this cultural unit is in some ways more foundational to human identity than even um a chimp social unit is for its identity um which I think it is then you know it no longer becomes an option so it kind of accelerates the epistemic uh ontology that I think the free principle um underwrites meeting kind of ser for self seeking information information gain so I think you could and I didn't but I think we can rephrase this as morality as information processing um so in in that context in the context of I'm a person I'm essentially a concept within a broader symbolic social reality um and it starts to I start to minimize uncertainty by um Computing cells by finding you know the the blanket that I'm behind um then morality in that case becomes a signal to what those identities are and where I belong so in that case um someone was deviant okay this person is evil then that's that's a big deal um let me pay attention to this um someone um you know yelled at me when I crossed the street because I was crossing the street at a wrong time that's going to stick in my mind because they've they've pegged me as something right so that was the use of morality as creating something um and so yeah so this is one of the the reasons morality is I think so Loom so large I think in in our minds um and I think is is important for for that check but yeah yeah you also have insight into you know these Concepts so yeah if there's the words I should be using please please stop me here yeah I think it's very interesting apply active inference to the yion or to the symbolic Journey to the search for meaning and and all of this so if we were just talking about like visual perception pixels on the screen or photons on the retina a sophisticated cognitive entity would do structure learning and propose latent causes like oh maybe it's not just a million red pixels moving from left to right maybe it's a red car and so then there's essential or the symbolic car so that would be like structure learning and World model again also for the kind of entities that we are it's like a selfin world model because one of the most important things is agency and understanding well what are the mov ments that I have agency over which is why we're able to kind of ignore is secades because we know or or suppress proprioception associated with our body being in the position that we expect it to be because those bodily features are under our control and so we're we're able to basically calibrate them out not need to be distracted or attend to them and then that allows us to be sensitized to changes that are not caused by us and so it's like what is the eye what is this symbolic and embodied strange Loop that's causing all my thoughts my thoughts and so it's like that can be framed very clearly in a uncertainty reduction structure learning external state

or hidden State framing and only um through kind of a lot of um modification and and coercion can that be cast in like a microeconomic reward maximization framework which just leaves um to be done than before the application and ACM is like well no this is at least a uncertainty reducing way approach that question of self and potentially multiscale self um Discovery yeah like in um yeah one of the the points I was trying to make with um neoclassical like similar to what Dar did in um in structuralism was well you because this this is such a this is a general equilibrium kind of I think of it as like a thin um framework meaning it um it reduces the economy down to um essentially get no movement because all the resources are allocated and there's really nothing nothing left to do um but the what's interesting is when people identify where that fails so you know classical market failures um it's the most interesting side of um you know classical economics it's it's how you get um pollution it's how you get technical spillovers that have you know positive reaction um I think an interesting one is in the concept of cultural niches uh social trust and so there the and in also the um and how firms Co see how firms don't exist so you'll have social trust at the level of um level of a country you'll have social trust at the level of the firm um and within that that minimizes uncertainty for the employees it minimizes uncertainty for the consumers because they know the product they're getting um and I think we can also think of it as minimizing uncertainty for um you think of someone who swell and they're not in the marketplace yet they're just learning things um so what's you know how do they participate um how can we think of them participating in the market economy other than just a um kind of a longtime consumer until they start um participating well they have they have a series of of of symbols and they have a series of ways of reducing their uncertainty um and so this is you know unions this is um universities this is you know companies that maybe they're not thinking in companies per se but they're thinking of um you know a business person or a scientist or you know they have these similar things that can be used for there tell another question was you described this kind of perspective swap that we engage in and you described it as there was a a perpendicular component to what what was that social setting that was perpendicular and what was perpendicular to what yeah yeah and so so here um you know so so so here I guess I'm kind of the on the edge of my not knowledge but what what I was the perpendicular would be uh so typical um primary relations but also um but also a lot of our social relations are are intimate and so that's perpendicular meaning in the sense of um you know for these are the people that you that you you know these are interactions that you have that are kind of based on um your subjective um perceptions and meaning those meaning those interactions that we can kind of

carve out of a symbolic reaction so for for I guess for for people it's it's difficult to do because as soon as you engage with someone um you know you're thinking about what you should be doing and and and you know there's a bunch of unconscious things of okay I should put you know I should shake their hand or you know I should um you know if I'm I should kiss their cheek or um you know I should take my shoes off now so there's obviously kind of a top down implicit um symbolic um occurrences happening however you know for non-human animals for for primates um you know for dos Etc um that world is you know it does have it does have history um you know it does have it does have memory but the perspective is still is still subjective it is still you know a subjective memory um there there's no kind of Above The Fray uh objective top down um sense for which they can coordinate their their actions and behaviors and their kind of goals um and so I on on one hand I don't think it's saying too much because I I think it's basically just describing um you it's kind of visualizing how symbols are different than um symbolic interactions are different than um um non-s symbolic interactions on the other hand I think it's interesting because uh I think it can help it can help us imagine the context and maybe model the context for how we first started it right how we we first started to do it um which is that we were in an environment where um both environment both in a in a um in an actual environment like in in the place but also our kind of cognition was was at a place where we could start to imagine our Behavior as if a third person were looking at it um and so that yeah that's essentially what I was what I was getting at cool all right I'm going to go to some questions from live chat so if you if you have any thoughts on them please feel free otherwise of course however you want to see them okay upcycle Club wrote how can tokenization improve the performance of active inference agents that use tensor networks as generative models for sequential data and then just followed up by defining tokenization is the process of breaking down a text into smaller units called tokens which can be words subwords characters or symbols do you have any thoughts on this repeat the question yeah yeah how can tokenization improve the performance of active inference agents that use tensor networks as generative models for sequential data you know I was just interested in in the question itself but yeah I have no I have no thoughts but other than I can't wait to you know to examine the the wording of that question and maybe have an opinion one day yeah appreciate appreciate whoever asked that whoever asked I thinking that I would have insights into that question indeed it's on stream we'll come back to it okay Dave asks a question there's a few background pieces but it will end with a question mark from what I see okay Whitehead in process in reality presents a framework that allows a kind of chaining across local spatial relations that

allows a group of observers to attribute objective locations to objects in a shared space a group of Minds can apparently share Concepts somewhat contrary to what we'd expect from a strict avoidance of the fallacy of misplaced concreteness it'd be nice to speak meaningfully of shared Concepts while avoiding the fallacy of misplaced concreteness and also avoid Whitehead's rather unsatisfactory Reliance on Eternal objects Concepts residing in the mind of God could the trick be made to work by positing chaining across locally I and thou first first and second person mappings where you and I take into account the partial discrepancies between what you and I mean by a given term or a given illusion while we retain and reuse the structural commonalities between your and my usage term yeah I mean I this this is certainly much more in my um in my wheelhouse than the in the first one um and yeah I mean I I start my paper talking about wiaad so I I certainly deserve a question that involves it um I I I think precisely and and I I think that that's kind of getting at what um you know the way I present kind of our Discovery in language we um approaching the idea of of you know the fallacy of misplaced concreteness like um we believe in these platonic forms it's this Western bias it's this thing that Plato did that was wrong that we continue to do with you know classical economics but but uh even though it's wrong it's it motivates us to create language because by believing something um by believing in these forms in a sense um we end up reifying we end up creating things that that do exist so I I guess the way I see it first of all is that um the myth of objectivities is well by wanting to believe that there is a right way to speak by wanting to believe that um there's a right thing there's a right thing I mean by the right thing um we end up kind of bringing that about um uh where I'm curious to kind of parse his question out is at the tail end where he talks about you know the difference the distance between I guess the two subjects I and you um and kind of how we kind of inch closer together I guess might simplify framework I'm thinking well if if both me and you believe that you know we're speaking more or less the same language um you know we're both talking about active inference like even even at at lower levels then we can navigate a closer and closer space and establish something like a reference frame where things where words have common meanings Etc um we can establish some common ground um and I think yeah so I think on one hand you can think of it as this if I understood correctly towards the end we can think of it as this kind of closing the gap between our two perspectives um but I think you also have to it's also necessary that we're both conceiving of this um you know I don't want to say Godlike because we were trying to avoid white head's use of Godlike but it Godlike is useful because um it if we just think of that there's a a that there's this fallacy and there's this um perspective that can see the

true form um so we in a sense we need this kind of Mythology I think um or we at least needed it in the past um to kind of get at something like language forms awesome all out a few a few notes on that and ask another question so in thinking about these appeals or abductive hypotheses about hidden causes it's like wow what is causing all of these horses I mean there's small ones and big ones and they have different patterns and some have four legs some have three legs one you know had this other number of legs one time so it's like what could be causing all of these sounds sights smells and there's basically two extreme structural hypotheses one of them is like the single like there's the that's the platonic horse it's like I don't even know what it is it's very different than any of the horses but it's just somehow is the structural cause it's the formal cause of all these Horses The Other Extreme structural hypothesis is like horses don't exist everything causes every like there's just sort of like an infinite component model now complexity is always found in between the one and the infinite and so that's where we actually do develop more well articulated accounts and what really does cause a horse to be a certain way in its accent in its real embodiment but we can see that kind of um monist or absolutist appeal as being like as you put a a good null hypothesis like potentially one of the best starting null hypothesis not going too deep into value judgments but I mean why not just start with saying maybe there's one thing and then we'll keep adding latent causes as we go but then to confuse the appeal to a single cause with the actuality of the single cause is kind of this challenging dance because if you take it as real then it's real as real and then to hold it at a distance though but it's hard to hold something at a distance when life and everything is on the line um and then one other piece what you said about like well the the the Godlike appeals of of um some of these like whiteheads formal approaches and you seem to express somewhat of an ambivalence about this kind of an appeal um and that reminded me of the quote from Hamlet what a piece of work is a man how Noble in reason how infinite in faculty in form and moving how Express and admirable in action how like an angel in apprehension how like a god action and inference in Hamlet early ACM days okay I'll ask a question from the chat Muse equal wrote isn't this indicative of learning an intuition being entropic postobservation we disassemble the observed construct into segment frames of historical time through interaction prior knowledge and memory where where is learning where is symbolic learning in what ways is symbolic learning anticipatory real time or retrospective I guess this kind um you know what you were saying with you know forms as kind of causes um of hidden States and kind of how do you um you know update you know update beliefs like once you have the lece like how complicated is it to um you know to

actually update them um trying to think of an example that that we went through um but I yeah it's a and now now I'm just thinking of of my daughter right because she's she's learning how to speak right so she's clearly symbolic learning um and you know kids are like much more comfortable um with variation um you know with with denture pre um I'm not comfortable but they they exist in much more entity than um than than we are um so they can they can navigate the space I think probably better than we can conceive that they that we can or that even one can um because they're in it so often um and how do you kind of build symbols well I guess um Okay so where I came from from um symbolic learning was um Hoffman and in Sanders's um um book on analogies I'm forgetting surface is in essence um and so their claim is that um which I think aligns pretty well with active inference um but I'm curious to hear more feedback on it um and their claim is that um we learned primarily through metaphor enemy um and so it's through I think through analogy they said so surf they look at we look at surface differences rather than essential differences and essential would be more form platonic form um look at more of surreal differences so um um you know my you know my dad is to me there's a dad towards someone else um you know I'm I'm crossing the street so I I I I suppose learning um I'm trying to think of some good examples but um you know we we build we build structures of uh symbols um using kind of like against like so you know this is like you know this is an action of sitting in in one chair um I sat I saw someone sitting on a bench outside um you know I saw someone sitting on the ground um so it's producing I guess if you a generative model of um you know in this context in this situation this is happening um and these are the um you know these are the relevant features of the environment um and this is then the object within that environment um here's something going on here and I'm bringing in and I'm using that as a prior to establish something um and then at some point and then at some point we we start getting Concepts and then once you have your initial concept because that takes some time once you have your initial concept then you compare Concepts to Concepts in this kind of metaphorical fashion I didn't give a good example of metaphor but um we start to move outward that way but yeah I guess that's the best that's the best I have for you at moment yeah this role of metaphor is very interesting and something lightly speculated upon but not quite brought into active inference generative modeling yet but metaphors is like well it's like the highest point on the building or it's like the center of the circle or it's like what's inside or outside of the circle so those would those kinds of metaphors which can be bootstrapped or grounded in in the immediacy of our Developmental and evolutionary history can then be brought into purely cognitive or symbolic spaces so we could talk about a clash of

symbols or about friction or inertia and grasping all of these tangible examples that are kind of out there but then we bring them in here and I I think that may have something to do with what Dave was discussing with the I and thou so it's not that's kind of the Martin boober it's not just I it if we were talking to um an inanimate object or an object that we didn't think had a selfin world model then it'd be like ranting to you know this puppet or just like kind of just but but under no expectation that it has a mind of its own but then the I thou or in active inference what's sometimes called like thinking through other minds or thinking with other Minds where it's kind of like well I'm modeling my own speech in relation implicitly or explicitly to what they'll respond to right and and knowing that they have that kind of recursive um activity as well that is purely cognitive it's it's lifted from the Accenture it's brought into the structural and the symbolic um but it's always realized by systems with accents and and I I I think just the way you introduced and and highlighted um that um direct experience of like how somebody is saying that not what they say it's easy for us now to focus on the semantics of the what which are so overwhelming but that almost obscures the kind of bootloader that actually got us to the developed social systems which as you pointed out clearly some things have changed such that we don't require that direct intimacy yeah and so that that's a a complex tangle of of cognitive and cultural phenomena it's not like only being ascribed to to the ability to do metaphor or think through other minds or use symbolic um logic and learning but all of these are are coming together in this kind of unified mystery yeah yeah yeah and um yeah maybe understand um that earlier point so the the frisen article on um singing um singing on from the same H sheet um or you think of it as you know with with language you're engaging in these um you know fast-paced Dances um with people you know exchanging um you know exchanging beliefs in anticipation of um you know your in interior beliefs um and you're Co co-constructing something um yeah I the only thing I want to say is uh when you ask earlier where I felt Act of inference was was useful I guess in a nutshell it can it can better understand the Dynamics between cooperation competition um you know outside and inside um and yeah that gets us some of the tension with how we think of communication whether it's Cooperative or whether it's kind of just ecological but it it encourages a um it encourages that Dynamic where we're we're in um like if you know if we're saying something and maybe I I'll disag maybe you'll disagree slightly with me on a point um but we're so we're still here trying to be cooperative um so we're trying to say things that we both understand um and sharing that kind of same cultural me space or attempting to and at the same time proposing either iteratives or modifications of those beliefs so that I'm able to persuade you of of my point um when we get

out whether communication is kind of Cooperative um to the extent that it's Cooperative to it's dance versus it's kind of um contentious then you can start to tease out those know I don't know those those factors okay short and interesting question in live chat upcycle Club writes what do you think about petroglyphs petroglyphs are rock carvings made by pecking directly on Rock surfaces using a stone chisel and a Hammerstone what do you think about rock inscription Rock inscription shely I'm big fan no U well I imagine you might be getting at is um uh is that uh so is that symbolic you know probably if human does it if um maybe he's getting it or she's getting it if a you know if if a bird has done it is that symbolic um yeah I don't know I'm curious but yeah I'm a big fan of of sculpture that kind of reminds me of the oam andus poem like I was here and all of that entails like there was an Empire here now of course in the poem it's all degraded and so that's kind of the media and the message but the inscription on the stone is like I was here I chose to do this or even if sorry what oh almost like graffiti or something like like why why are we why are we doing this why are we okay I get it yeah somebody somebody who uses that graffiti tag did that we could have different stories they could have different stories about whether they have free will or all these different like secondary things but it's like the fact of the matter is on the rock it was inscribed so I I I say yes so I mean I I get I get that now so um what I see there is um you know what I see there is that is you know that's soul right that that's uh in in in the sense of um um you know something like like Whitman would would describe it is that that is um we we do that because we have a soul ademy not that in a um not necessarily in a religious context but um as in you can exist uh you know daniel Freeman can exist as a conceptual identity outside of space and time um and the way you do that is that you you leave a mark you know you leave your legacy and um you know um you know etc etc but yeah so we we exist as these symbolic selves both um as part of these cultural units but in some sense outside of it um and so yeah do you write your name on a wall do you um you know do you start a company um you know do you want to be um you know famous do you want to be on a movie for whatever reason um yeah this is that that is an extension of identity um yes so if that's what that's getting at then yeah um cool well in sort of closing where are you going to take this line after the Frontiers paper and this what's the next Frontier or the next Journey Al yes so that that that I don't know so yeah I'm open to ideas but um yeah yeah I was I was obsessed with this idea for a while um and I think I've nearly gotten it I think I'm kind of like 95% the way there or maybe 90% um and I think this discussion has helped me but yeah I'm just kind of curious to see um where it might go but yeah I mean I'd love to see it formalized love to see it taken seriously but um yeah have no

specific uh has no specifics but cool it was a great with you I appreciate the setting us up and I appreciate the dialogue thank you I really appreciate hearing the comprehensiv and just the relatability of the Curiosity and how many things are brought together in your paper and in this discussion so see you around till next time all right take care thanks everyone bye